Volume 03 - Book 03.19 Infrastructure

Version v21 draft

README.md

Book 03.19 - Infrastructure

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-README

Purpose

Define the Infrastructure blueprint package and its subsystem decomposition as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for the Infrastructure blueprint package and its subsystem decomposition. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for the Infrastructure blueprint package and its subsystem decomposition.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for the Infrastructure blueprint package and its subsystem decomposition.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.

- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- InfrastructureDefined
- InfrastructureRegistered
- InfrastructureStateChanged
- InfrastructureReviewRequested
- InfrastructureGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.

- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Infrastructure has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19 - Infrastructure

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-INDEX

Purpose

Define the Infrastructure blueprint package and its subsystem decomposition as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for the Infrastructure blueprint package and its subsystem decomposition. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

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- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for the Infrastructure blueprint package and its subsystem decomposition.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

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- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Infrastructure has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
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Traceability

- MAL-V00-B009
- MAL-V00-B010
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- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19.01 - Infrastructure Overview

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-001

Purpose

Define Infrastructure Overview within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Infrastructure Overview within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Infrastructure Overview within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Infrastructure Overview within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- InfrastructureOverviewDefined
- InfrastructureOverviewRegistered
- InfrastructureOverviewStateChanged
- InfrastructureOverviewReviewRequested
- InfrastructureOverviewGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Infrastructure Overview has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19.02 - Environment Model

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-002

Purpose

Define Environment Model within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Environment Model within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Environment Model within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Environment Model within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- EnvironmentModelDefined
- EnvironmentModelRegistered
- EnvironmentModelStateChanged
- EnvironmentModelReviewRequested
- EnvironmentModelGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Environment Model has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19.03 - Deployment Pipeline

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-003

Purpose

Define Deployment Pipeline within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Deployment Pipeline within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Deployment Pipeline within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Deployment Pipeline within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- DeploymentPipelineDefined
- DeploymentPipelineRegistered
- DeploymentPipelineStateChanged
- DeploymentPipelineReviewRequested
- DeploymentPipelineGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Deployment Pipeline has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

book-03-19-04-storage-layer.md

Book 03.19.04 - Storage Layer

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-004

Purpose

Define Storage Layer within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Storage Layer within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Storage Layer within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Storage Layer within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- StorageLayerDefined
- StorageLayerRegistered
- StorageLayerStateChanged
- StorageLayerReviewRequested
- StorageLayerGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
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Security

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- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Storage Layer has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

book-03-19-05-cache-layer.md

Book 03.19.05 - Cache Layer

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-005

Purpose

Define Cache Layer within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Cache Layer within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Cache Layer within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Cache Layer within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- CacheLayerDefined
- CacheLayerRegistered
- CacheLayerStateChanged
- CacheLayerReviewRequested
- CacheLayerGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Cache Layer has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19.06 - Networking

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-006

Purpose

Define Networking within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Networking within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Networking within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Networking within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- NetworkingDefined
- NetworkingRegistered
- NetworkingStateChanged
- NetworkingReviewRequested
- NetworkingGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Networking has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19.07 - Secrets Management

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-007

Purpose

Define Secrets Management within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Secrets Management within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Secrets Management within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Secrets Management within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- SecretsManagementDefined
- SecretsManagementRegistered
- SecretsManagementStateChanged
- SecretsManagementReviewRequested
- SecretsManagementGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Secrets Management has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

book-03-19-08-sandboxing.md

Book 03.19.08 - Sandboxing

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-008

Purpose

Define Sandboxing within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Sandboxing within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Sandboxing within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Sandboxing within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- SandboxingDefined
- SandboxingRegistered
- SandboxingStateChanged
- SandboxingReviewRequested
- SandboxingGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Sandboxing has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19.09 - Metrics

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-009

Purpose

Define Metrics within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Metrics within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Metrics within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Metrics within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- MetricsDefined
- MetricsRegistered
- MetricsStateChanged
- MetricsReviewRequested
- MetricsGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Metrics has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19.10 - Tracing

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-010

Purpose

Define Tracing within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Tracing within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Tracing within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Tracing within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- TracingDefined
- TracingRegistered
- TracingStateChanged
- TracingReviewRequested
- TracingGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Tracing has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19.11 - Logs

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-011

Purpose

Define Logs within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Logs within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Logs within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Logs within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.
- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.

- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- LogsDefined
- LogsRegistered
- LogsStateChanged
- LogsReviewRequested
- LogsGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.
- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Logs has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

book-03-19-12-monitoring.md

Book 03.19.12 - Monitoring

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-012

Purpose

Define Monitoring within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Monitoring within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Monitoring within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Monitoring within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- MonitoringDefined
- MonitoringRegistered
- MonitoringStateChanged
- MonitoringReviewRequested
- MonitoringGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Monitoring has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19.13 - Backup Recovery

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-013

Purpose

Define Backup Recovery within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Backup Recovery within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Backup Recovery within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Backup Recovery within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- BackupRecoveryDefined
- BackupRecoveryRegistered
- BackupRecoveryStateChanged
- BackupRecoveryReviewRequested
- BackupRecoveryGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Backup Recovery has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

book-03-19-14-scaling.md

Book 03.19.14 - Scaling

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-014

Purpose

Define Scaling within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Scaling within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Scaling within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Scaling within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- ScalingDefined
- ScalingRegistered
- ScalingStateChanged
- ScalingReviewRequested
- ScalingGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Scaling has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19.15 - Infrastructure Governance

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-015

Purpose

Define Infrastructure Governance within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Infrastructure Governance within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Infrastructure Governance within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Infrastructure Governance within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- InfrastructureGovernanceDefined
- InfrastructureGovernanceRegistered
- InfrastructureGovernanceStateChanged
- InfrastructureGovernanceReviewRequested
- InfrastructureGovernanceGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Infrastructure Governance has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19.16 - Infrastructure Events

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-016

Purpose

Define Infrastructure Events within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Infrastructure Events within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Infrastructure Events within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Infrastructure Events within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- InfrastructureEventsDefined
- InfrastructureEventsRegistered
- InfrastructureEventsStateChanged
- InfrastructureEventsReviewRequested
- InfrastructureEventsGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Infrastructure Events has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19.17 - Infrastructure Storage

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-017

Purpose

Define Infrastructure Storage within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Infrastructure Storage within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Infrastructure Storage within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Infrastructure Storage within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- InfrastructureStorageDefined
- InfrastructureStorageRegistered
- InfrastructureStorageStateChanged
- InfrastructureStorageReviewRequested
- InfrastructureStorageGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Infrastructure Storage has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

book-03-19-18-infrastructure-security.md

Book 03.19.18 - Infrastructure Security

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-018

Purpose

Define Infrastructure Security within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Infrastructure Security within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Infrastructure Security within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Infrastructure Security within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- InfrastructureSecurityDefined
- InfrastructureSecurityRegistered
- InfrastructureSecurityStateChanged
- InfrastructureSecurityReviewRequested
- InfrastructureSecurityGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Infrastructure Security has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

book-03-19-19-infrastructure-testing.md

Book 03.19.19 - Infrastructure Testing

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-019

Purpose

Define Infrastructure Testing within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Infrastructure Testing within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Infrastructure Testing within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Infrastructure Testing within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- InfrastructureTestingDefined
- InfrastructureTestingRegistered
- InfrastructureTestingStateChanged
- InfrastructureTestingReviewRequested
- InfrastructureTestingGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

Security

- Apply least privilege to users, agents, services, providers, plugins, connectors, runtimes, and storage records.
- Deny protected actions when required permissions, policy checks, evidence, or human review gates are missing.
- Classify inputs, outputs, events, logs, metrics, memory, and exported artifacts before sharing or persistence.
- Audit privileged actions, denied actions, governance exceptions, configuration changes, and lifecycle transitions.
- Prevent generated or inferred outputs from becoming authoritative without explicit acceptance rules.

Metrics

- Adoption rate, successful completion rate, rejection rate, rework count, and user trust indicators.
- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
- Quality score, benchmark pass rate, validation pass rate, review backlog, and stale-record count.
- Security denial count, governance exception count, policy violation count, and audit completeness.

Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Infrastructure Testing has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
- Governance, security, and review controls are visible and traceable.
- Release artifacts include this Markdown file, the book PDF, MANIFEST.json, release notes, and the versioned ZIP.

Traceability

- MAL-V00-B009
- MAL-V00-B010
- MAL-V00-B011
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- MAL-V02-B019
- MAL-V03-B03-01-004
- MAL-V03-B03-05-003

Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.

Book 03.19.20 - Infrastructure Roadmap

Version: v21 draft

Status: Draft

Document ID: MAL-V03-B03-19-020

Purpose

Define Infrastructure Roadmap within the Infrastructure blueprint as part of Volume 03 Master Blueprint for Mariam AI Enterprise OS.

Scope

This document covers architecture-level responsibilities, contracts, interfaces, events, storage expectations, security controls, metrics, testing, acceptance criteria, and traceability for Infrastructure Roadmap within the Infrastructure blueprint. It does not implement product code or bypass the documentation-first governance model.

Responsibilities

- Establish a canonical blueprint boundary for Infrastructure Roadmap within the Infrastructure blueprint.
- Convert strategic roadmap intent into implementation-ready subsystem contracts.
- Coordinate with Enterprise Core identity, runtime, event, storage, security, and observability services.
- Preserve governance, traceability, reviewability, and rollback expectations for high-impact behavior.
- Provide enough detail for future specifications, testing guides, operations guides, and implementation issues.

Inputs

- Volume 00 Enterprise Constitution principles for governance, security, quality, engineering, AI, DNA, swarm, and evolution.
- Volume 01 Master Vision strategy for enterprise outcomes, knowledge leverage, marketplace expansion, and autonomous work.
- Volume 02 Master Roadmap phase guidance corresponding to Infrastructure.
- Earlier Volume 03 blueprint books for Enterprise Core, Enterprise Organization, AI Society, Knowledge System, and Capability System where dependencies apply.
- Domain requirements, user intent, policy context, runtime constraints, and release acceptance evidence.

Outputs

- Blueprint contract and subsystem boundaries for Infrastructure Roadmap within the Infrastructure blueprint.
- Canonical records, events, metrics, storage expectations, and security controls.
- Traceable inputs for Volume 06 specifications and later specialist volumes.
- Acceptance evidence that downstream implementation can use before writing system code.

Interfaces

- Enterprise Core kernel, runtime manager, event bus, object manager, identity access, storage, security, and observability interfaces.

- Enterprise Organization decision, approval, escalation, KPI, and operating rhythm interfaces.
- AI Society agent, planning, execution, review, validation, and governance interfaces.
- Knowledge System graph, store, search, indexing, provenance, and quality interfaces.
- Capability System registry, matching, lifecycle, benchmark, governance, and execution interfaces.

Events

- InfrastructureRoadmapDefined
- InfrastructureRoadmapRegistered
- InfrastructureRoadmapStateChanged
- InfrastructureRoadmapReviewRequested
- InfrastructureRoadmapGovernanceBlocked

Storage

- Canonical records for definitions, versions, owners, state, policy classification, and dependency metadata.
- Operational journals for lifecycle changes, execution evidence, review outcomes, exceptions, and acceptance decisions.
- Audit metadata linking user intent, agent or service identity, source context, runtime, model or provider choice, and policy status.
- Retention metadata that separates draft, active, archived, deprecated, and superseded records.

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Metrics

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- Latency, cost, throughput, availability, error rate, fallback rate, and recovery time.
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Testing

- Contract tests for required inputs, outputs, interfaces, event envelopes, storage records, and lifecycle states.
- Security tests for tenant isolation, least privilege, protected action denial, redaction, and audit integrity.
- Quality tests for correctness, repeatability, ranking or routing behavior, recovery paths, and acceptance evidence.
- Integration tests across Enterprise Core, Organization, AI Society, Knowledge System, and Capability System dependencies.

- Release tests verifying Markdown inclusion, PDF generation, ZIP packaging, manifest checksums, and release notes.

Acceptance Criteria

- Infrastructure Roadmap has explicit ownership, lifecycle, interfaces, events, storage, security, metrics, and testing expectations.
- The blueprint is detailed enough for downstream specifications without requiring implementation code in this repository.
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Version History

Version	Date	Status	Notes
v21 draft	2026-06-29	Draft	Initial Infrastructure blueprint content for Mariam Architecture Library v21.